

DOCUMENT 03

Advanced Airframe & Propulsion Research

Laminar Fuselage · Rotor Wake · Structural Batteries · ANC · Phase 3 Roadmap

Phase-2 and Phase-3 advanced airframe R&D; beyond the validated Tarot TL65B44 test platform. Covers CFD methodology for the laminar teardrop fuselage, rotor vertical-offset optimisation derived from DLR 2025 data, structural battery feasibility analysis, acoustic cancellation DSP design, variable-diameter propeller concept, and the complete Phase 3 research roadmap. Every concept is rated by TRL, estimated gain, and implementation risk.

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1. RESEARCH PROGRAMME CONTEXT

All acoustic, vibration, structural, and endurance measurements in the core paper were conducted on a standard Tarot TL65B44 CFRP X-quad frame — a symmetric, boxy quadrotor geometry with no aerodynamic shaping. The advanced airframe concepts in this document are research targets for Phase 2 (laminar fuselage, rotor offset) and Phase 3 (structural batteries, acoustic cancellation). The distinction between validated results and research targets is maintained throughout.

The research programme is deliberately sequenced from highest return-on-investment to lowest. Rotor vertical offset provides measurable efficiency gain with minimal hardware change and is designated Phase 1. The laminar fuselage requires CFD validation and new manufacturing before building, making it Phase 2. Structural batteries and acoustic cancellation require materials advances and have lower near-term priority.

Concept	TRL	Phase	Gain Type	Predicted Gain	Key Risk
Rotor vertical offset z/D=0.35	3–4	1	Hover efficiency	+7.7% rear rotor thrust	Minor mass penalty from mounts
Laminar teardrop fuselage	2–3	2	Cruise drag reduction	8–15% cruise power reduction	Rotor turbulence disrupts laminar regime at hover
Airfoil-section arm tubes	3	2	Cruise and hover drag	+7.6% endurance (Freitas 2025)	Speed reduction –8.6%
Structural battery CFRP arms	2–3	3	Energy density	+15–25% capacity at same mass	Safety (crash = thermal event); low cell energy density
Variable-diameter centrifugal props	1–2	3	Hover efficiency	+15–25% induced power reduction	Blade structural integrity at >550N centrifugal load
Active acoustic cancellation	2	3	Noise floor	+5–8 dB beyond FOC	Latency budget <1 ms; marginal vs FOC gain
Morphing arm geometry	1–2	Deferred	Cruise efficiency	<5% at 10–15 m/s	Actuator mass >efficiency gain at 1.5 kg scale
Push-flow propulsion	1	Rejected	—	Negative at 1.5 kg	No physical mechanism at this scale

Table 1. Advanced airframe concept assessment matrix.

2. ROTOR VERTICAL OFFSET OPTIMISATION

2.1 Physical Mechanism

In a standard flat-plane quadrotor, the front rotor pair generates a turbulent wake that propagates rearward and downward. The rear rotor pair ingests this wake, experiencing increased effective angle-of-attack variation, elevated profile drag, and reduced thrust efficiency. DLR (German Aerospace Centre) 2025 experimental data on a representative quadrotor platform demonstrated that raising rear rotors by $z/D = 0.30\text{--}0.40$ (where D is rotor diameter) moves them above the front-rotor wake plane, recovering 7.7% of the thrust that would otherwise be lost to wake ingestion.

For the target platform with $D = 228.6\text{ mm}$ (9-inch props), the optimal vertical offset is:

$$\Delta z_{\text{opt}} = 0.35 \times 228.6\text{ mm} = 80\text{ mm}$$

This is achievable by machining stepped arm mounts that position the rear arm pair 80 mm higher than the front pair. The mass addition is approximately 15 g of aluminium mounting hardware — negligible relative to the efficiency gain. The geometry change also reduces inter-rotor acoustic coupling, producing an estimated additional 1–2 dB SPL reduction at the rear blade passage frequency harmonics.

2.2 Predicted Performance vs Flat Configuration

Parameter	Flat X-Quad (Current)	z/D=0.20	z/D=0.35 (Optimal)	z/D=0.50
Rear-rotor thrust efficiency	100% baseline	+3.5%	+7.7%	+5.2% (diminishing)
System hover power (no payload)	36.1 W	34.8 W	33.7 W	34.4 W
Predicted hover endurance	65 min	67 min	70 min	68 min
Acoustic coupling term (rear)	0 dB baseline	−0.5 dB	−1.8 dB	−2.5 dB
Mount mass penalty	0 g	+10 g	+15 g	+22 g
CG shift (rear rotors higher)	None	+2 mm	+3.5 mm	+5 mm

Table 2. Rotor offset optimisation predictions (no payload, validated k_empirical).

2.3 Implementation Plan

- Fabricate stepped CFRP arm mounts: CNC from 6 mm Al 6061 plate; anodise black; M4 through-bolts
- Recalibrate flight controller: update EKF2_EV_POS_X and motor geometry parameters to reflect offset geometry
- Thrust stand validation: compare rear-rotor thrust at identical motor current with flat vs offset config (n=10 per condition)
- Success criterion: ≥5% thrust increase from rear pair at identical electrical input
- If validated: design offset mount as standard for all Phase 1 drones; update BOM with \$15 cost addition

3. LAMINAR TEARDROP FUSELAGE — FULL CFD PLAN

3.1 Design Rationale and Geometry

The current Tarot TL65B44 has a boxy central plate body with circular-section CFRP arm tubes. The proposed teardrop fuselage encloses all central avionics in a streamlined 2:1 length-to-width body inspired by NACA symmetric airfoil cross-sections. The frontal area reduces from approximately 280 cm² to 180 cm² — a 36% reduction — and the drag coefficient decreases from Cd ≈ 1.0 to Cd ≈ 0.65–0.80. The drag power term in forward flight:

$$P_{drag} = \frac{1}{2} \times \rho \times C_d \times A_f \times V^3$$

scales with the cube of airspeed. At V = 10 m/s, the drag power reduction from the teardrop fuselage is: (0.70 × 0.018) vs (1.0 × 0.028) = a 55% reduction in aerodynamic drag power. At cruise, this directly extends range. At hover, the fuselage body is inside the rotor-induced downwash column and boundary layer behaviour is dominated by turbulent rotor wake — laminar flow over the fuselage is not achievable in hover, and the drag benefit at hover is negligible.

Critical Design Constraint — Hover vs Cruise

- Laminar fuselage drag reduction applies ONLY during forward flight (V > 8 m/s)
- At hover: rotor-induced turbulence (Re ~10■) prevents sustained laminar boundary layer on fuselage
- Mission type matters: for range-limited transit missions → high gain; for stationary surveillance hover → minimal gain
- The drag reduction claim is a CRUISE endurance benefit, never a hover endurance benefit
- Aerodynamic priority ranking: (1) rotor offset, (2) airfoil arms, (3) laminar fuselage — in order of implementation

3.2 Reynolds Number Analysis

The critical aerodynamic parameter is the Reynolds number at the fuselage:

$$Re = \rho \cdot V \cdot L / \mu = 1.225 \times V \times 0.38 / 1.81 \times 10^{-4} = 25,800 \times V$$

At V = 10 m/s: Re = 258,000. At V = 15 m/s: Re = 387,000. This is the critical transitional regime (Re 10⁵–10⁶) where laminar-to-turbulent boundary layer transition occurs and surface quality is paramount. Required surface finish Ra < 50 µm for sustained laminar flow, achievable with CFRP epoxy layup on a polished mould. The docking keel protrusion on the fuselage belly creates a local separation bubble; CFD must quantify the keel drag penalty and verify it remains below the 2% threshold.

3.3 Full CFD Simulation Matrix

Case ID	Geometry	Condition	AoA	Speed	Re	Output
C01	Tarot boxy (baseline)	Clean cruise	5°	10 m/s	210k	Cd, CL, pressure distribution
C02	Teardrop (no keel)	Clean cruise	5°	10 m/s	210k	Cd, CL — compare vs C01
C03	Teardrop + keel rail	Clean cruise	5°	10 m/s	210k	Keel drag penalty %
C04	Teardrop + keel	Clean cruise	5°	15 m/s	387k	Cd at higher Re
C05	Teardrop + keel	Hover (V=0)	0°	0 m/s	—	Hover drag in rotor downwash
C06	Airfoil arm tubes	Clean cruise	5°	10 m/s	210k	Arm drag vs circular section
C07	Full config (fuselage+airfoil arms)	Clean cruise	5°	10 m/s	210k	System Cd, L/D
C08	Full config	Gust (side wind)	0°	8 m/s lateral	167k	Lateral force during docking
C09	Teardrop + keel	Transition study	5°	6 m/s	155k	Laminar separation bubble location

Table 3. CFD simulation matrix for laminar fuselage validation.

3.4 CFD Methodology Details

Solver: ANSYS Fluent 2024 R1 with RANS k-ω SST turbulence model — appropriate for transitional flows at Re 10⁵–10⁶. Transition sensitivity studied via γ-Reθ transition model.

Mesh strategy: Structured inflation layers on all solid surfaces; first cell y+ < 1 for accurate near-wall resolution; 15 inflation layers with 1.2 growth ratio; unstructured tetrahedral in far field. Total cell count: 8–12 million per case. Mesh independence verified by comparing 4M vs 8M vs 12M cell results — converge within 1%.

Boundary conditions: Velocity inlet at domain entry; pressure outlet at exit; no-slip wall on all solid surfaces; symmetry planes where geometry allows. Far-field dimensions: 20 × fuselage length in all directions to avoid wall confinement effects.

Convergence criteria: Residuals < 10⁻⁶ for all equations; lift and drag coefficients stable over last 200 iterations (variation < 0.1%).

Validation: Before running production cases, validate solver against published NACA 0018 wind-tunnel data at Re 200,000 (available from NASA/NACA technical reports). Accept if predicted Cd within ±5% of published data.

Post-processing: Extract integrated Cd, CL, L/D for each case; plot wall shear stress to identify laminar/turbulent transition location; compare boundary layer displacement thickness at fuselage mid-section; visualise pressure coefficient Cp distribution.

3.5 Manufacturing Specification for Prototype

- Material: T700/Epoxy CFRP, [0/±45/90]s layup, 60% fibre volume fraction

- Mould: CNC-machined aluminium female mould from frozen CFD geometry; polished to Ra < 25 µm
- Process: Wet layup or vacuum infusion; autoclave cure at 120°C/6 bar recommended for quality
- Surface finish: Post-cure wet-sand to Ra < 50 µm; apply automotive-grade clear coat
- Integration: Internal battery bay and avionics mounting designed to allow fuselage shell to be split horizontally for access; magnetic latches hold shell halves together
- Docking keel: Integral to lower shell half; machined from same CFRP layup; keel slot accepts aluminium keel rail insert with M3 threaded inserts
- Target mass: ≤350 g for complete fuselage shell (vs ≈120 g for current flat plates — penalty accepted for drag benefit)

4. STRUCTURAL BATTERY — FEASIBILITY DEEP DIVE

4.1 Technology Readiness and Energy Density

True structural battery composites use carbon fibres as electrodes (negative) and structural glass fibres coated with LiFePO4 as the positive electrode, with a structural polymer electrolyte providing both ion transport and mechanical load transfer. This architecture is being developed by Chalmers University and has demonstrated specific energy of 24 Wh/kg at the cell level in research prototypes (Asp et al., 2021). Commercial structural battery products aimed at small UAVs (Kühnelt et al., 2022) achieve 52–100 Wh/kg at the structural element level.

By comparison, conventional Li-ion 18650 cells achieve 200–250 Wh/kg at cell level and 150–180 Wh/kg at pack level. Structural batteries currently deliver 1/3 to 1/2 of the energy density of conventional packs while providing structural function.

System	Energy Density	Structural Function	Safety Record	TRL	Cost/W h
Li-ion 18650 (conventional)	200–250 Wh/kg cell	None	Mature BMS protocols	9	~\$0.10–0.20
LiPo pouch (conventional)	150–200 Wh/kg cell	None	Mature (fragile)	9	~\$0.15–0.30
Structural battery (research)	24–100 Wh/kg element	Yes — significant	Limited data	3–4	~\$5–20
Embedded pouch in CFRP (practical)	100–130 Wh/kg system	Partial — reduced E	Unknown under crash	3	~\$1–3

Table 4. Structural battery technology comparison.

4.2 Break-Even Analysis

For the structural battery concept to provide net energy benefit over the current configuration:

Let M_arm = 128 g (current CFRP 4-arm set)

If converted to structural battery elements at 100 Wh/kg:

E_structural = 0.128 kg × 100 Wh/kg = 12.8 Wh additional energy

Current battery: 148 Wh (W1 Li-ion). Addition: +8.6% capacity.

But: structural arm elements have reduced stiffness vs pure CFRP.

Safety factor drops from 4.83 to estimated 2.8–3.2 — still above 2.5 minimum.

Mass increase vs conventional CFRP arms: +0 g (same arm mass, different material function).

The break-even is favourable in terms of energy (+8.6% at zero mass cost) but the structural safety reduction, thermal runaway risk in a crash, and current TRL 3 status make this unsuitable for Phase 1 or Phase 2 deployment.

The decision rule: revisit when structural batteries reach TRL 6 with validated crash thermal containment.

4.3 Practical Near-Term Approach

A practical intermediate step is to embed standard LiPo pouch cells inside hollow CFRP arm tubes, with the arm tube providing both structural function and battery containment. This differs from true structural batteries: the cell provides zero structural contribution, and the arm provides zero electrochemical function. Benefits: (a) no degradation of arm structural properties; (b) cells tested and characterised independently before integration; (c) cell replacement possible without arm replacement. Mass budget: hollow arm tube volume $\approx 15\text{ cm}^3$ per arm; at 600 Wh/L for LiPo pouch cells, each arm can hold $\sim 9\text{ Wh} \times 4\text{ arms} = 36\text{ Wh}$ additional, a 24% increase over the W1 battery.

Embedded Arm Cell Safety Protocol

- Use prismatic LiFePO4 cells only (thermal runaway temperature $>270^\circ\text{C}$ vs 150°C for NMC) in arm tubes
- Install fusible link between arm cell and main bus: disconnect on crash detection (accelerometer $>20\text{g}$)
- Arm tube must maintain structural integrity at cell temperature up to 200°C (CFRP decomposes at $\sim 300^\circ\text{C}$)
- Post-crash inspection mandatory: visual + thermal camera scan of all arms before reuse
- Phase 3 only: implement only after 100-cycle embedded cell bench characterisation

5. ACTIVE ACOUSTIC NOISE CANCELLATION — DSP DESIGN

5.1 Signal Chain and Latency Budget

Active Noise Cancellation (ANC) for propeller noise uses an adaptive filter (Filtered-x LMS) running on an embedded DSP to generate an anti-phase acoustic signal from a speaker array. The fundamental constraint is latency: the system must generate the cancellation signal within one period of the highest target frequency. For cancellation up to 1,000 Hz:

$T_{\text{period}} = 1/1,000\text{ Hz} = 1.0\text{ ms}$ — total DSP latency budget

Stage	Component	Latency	Cumulative
ADC conversion	STM32H7 HAL ADC at 192 kSPS	5.2 μs	5.2 μs
Anti-aliasing filter	8-tap FIR at 192 kHz	0.2 μs	5.4 μs
FxLMS filter (512 tap)	STM32H7 DSP instructions at 480 MHz	53 μs	58.4 μs
DAC output	STM32H7 HAL DAC at 192 kSPS	5.2 μs	63.6 μs
Amplifier slew	Class-D amp (PAM8403)	10 μs	73.6 μs
Acoustic propagation	0.5 m at 343 m/s	1,460 μs	1,534 μs ← EXCEEDS BUDGET

Table 5. ANC latency budget — acoustic propagation path dominates.

The acoustic propagation delay (1.46 ms for a 0.5 m path) exceeds the 1.0 ms latency budget. This is the fundamental constraint on ANC for drone propellers at 1 m separation. The solution is to use a feedforward architecture with a reference microphone close to the propeller source (minimising propagation delay to the reference signal) and a cancellation speaker at the target zone. For a 0.1 m path to reference mic: propagation = 0.29 ms, bringing total latency to 0.36 ms — within budget for up to 2,700 Hz cancellation. Practically achievable: 5–8 dB reduction at BPF harmonics (167–500 Hz for 5,000 RPM, 2-blade props).

5.2 Hardware Specification for ANC Experiment

Component	Specification	Mass	Power	Cost
Reference microphone	MEMS, 20 Hz–10 kHz, 66 dB SNR, arm-mounted	1 g	0.5 mW	\$8
Error microphone	Same spec, at target zone	1 g	0.5 mW	\$8
DSP board	STM32H750 on custom PCB, 480 MHz, FPU	12 g	0.8 W	\$25
Amplifier	PAM8403 class-D, 3W×2, 90% efficiency	5 g	1.5 W peak	\$4
Speaker	Custom wound neodymium, 50 mm dia, 4Ω	28 g	3 W	\$12
Total ANC system	Full feedforward FxLMS	47 g	~4.8 W	\$57

Table 6. ANC experiment hardware specification.

Power cost assessment: 4.8 W for ANC system on a drone with 42 W hover power = 11.4% power increase. At $W^{1.5}$ scaling, this mass addition (+47 g) and power draw would reduce endurance by approximately 8–12%. The ANC system must achieve >10 dB reduction to justify its power and mass cost. Based on the latency analysis, practical cancellation is 5–8 dB — likely insufficient to justify deployment. FOC already achieves 13 dB at zero mass and power cost. Verdict: Phase 3 research experiment only.

6. PHASE 3 RESEARCH ROADMAP

Activity	Phase	Duration	Prerequisite	Success Criterion
Rotor offset mount fabrication + thrust stand validation	1	1 month	Phase 1 HW	≥5% rear-pair thrust increase
Airfoil arm tube design + wind-tunnel drag	2	2 months	ANSYS access	<0.75 Cd at 10 m/s
Teardrop fuselage CFD (9 cases)	2	2–3 months	ANSYS Fluent	Cd < 0.75 at cruise; keel penalty < 2%
Teardrop fuselage prototype fabrication	2	3–4 months	CFD frozen	Mass ≤ 350 g; Ra < 50 μm
Flight drag measurement (VICON force plate)	2	1 month	Prototype built	≥40% drag reduction vs boxy at 10 m/s
Embedded LiFePO4 arm cell bench test	3	3–4 months	Phase 2 complete	≥80% rated capacity after 100 cycles; no thermal events
ANC experiment (anechoic chamber)	3	2 months	Phase 1 HW	Measure dB reduction; compare vs FOC baseline
Variable-diameter prop prototype	3	6 months	Materials partner	≥15% hover power reduction vs fixed props

Table 7. Phase 3 advanced airframe research roadmap.

7. DEFERRED AND REJECTED CONCEPTS

7.1 Morphing Arm Geometry — Deferred

Variable-geometry arms (telescoping or tilting) were evaluated as a potential efficiency improvement for cruise flight. The aerodynamic benefit at 10–15 m/s for a 1.5 kg drone is estimated at 2–5% power reduction — marginal. The implementation cost is approximately 100–200 g of actuator hardware per arm (servo + linkage + bearings), totalling 400–800 g of mechanism mass on a 1.65 kg drone. This represents 24–48% of total drone mass added for a 2–5% efficiency gain. The mass penalty would increase hover power by more than the efficiency gain saves. Decision: defer

indefinitely; revisit only if drone class scales to >5 kg where actuator mass fraction becomes acceptable.

7.2 Push-Flow Propulsion — Rejected

Push-flow or flow-through propulsion proposes routing rotor airflow through or under the fuselage to reduce aerodynamic interference. The concept is physically invalid at the 1.5 kg drone scale for three reasons: (1) at typical hover and cruise speeds ($V < 20$ m/s), there is insufficient ram pressure for turbofan-style flow integration; (2) any internal duct increases weight and blockage drag more than it recovers from pressure recovery; (3) inlet and outlet geometry required for net positive efficiency is incompatible with the omnidirectional hover requirement of a multirotor platform. No CFD study is planned. Decision: permanently rejected.